

Curriculum Vitae — Glenn N. Heistand, P.E., CFM, BC.WRE

DRAFT FOR REVIEW — PRI Promotion Package 2027

⚠ Items flagged [NEEDS INPUT] should be confirmed or expanded by Glenn before final submission.

Glenn N. Heistand, P.E., CFM, BC.WRE
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ORCID: [NEEDS INPUT — provide ORCID if applicable]

1. Experience and Professional Development

Educational Background

Degree	Institution	Year
M.S., Environmental Engineering (Civil Engineering)	University of Illinois at Urbana-Champaign	2017
B.S., Civil Engineering	Ohio Northern University	1995

Professional Experience

Head, Coordinated Hazard Assessment and Mapping Program (CHAMP) Prairie Research Institute – Illinois State Water Survey (ISWS), University of Illinois at Urbana-Champaign *April 2021 – Present*

- Serves as Principal Investigator and section head for Illinois' FEMA Cooperating Technical Partners (CTP) program
- Administers approximately \$15.4 million in active FEMA/DHS cooperative agreements and state contracts
- Directs a 27-person multidisciplinary team of engineers, GIS specialists, project managers, and outreach professionals
- Leads strategic planning, staff development, federal reporting, and programmatic innovation for CHAMP
- Drives AI integration through the CHAMP AI Workgroup; developed custom GPT expert systems and Python automation that reduced quarterly reporting time by ~50%

- Member, ISWS Senior Leadership Team
- Affiliate Researcher, Water Resources Engineering and Science (WRES), Grainger College of Engineering, UIUC

Senior Hydraulic Engineer, CHAMP Illinois State Water Survey – Prairie Research Institute, UIUC *November 2009 – April 2021*

- Served as technical lead for regulatory flood studies across 25+ Illinois counties
- Directed H&H modeling, FEMA compliance review, and flood map production
- Mentored staff on advanced 1D/2D HEC-RAS modeling and FEMA technical standards
- Represented CHAMP at FEMA, IDNR, and interagency meetings
- Contributed to growth and national reputation of CHAMP program

Water Resources Engineer EMH&T (Engineers, Mechanical, Hydraulics & Technicians), Columbus, OH *May 1995 – October 2009*

- Designed over 100 stormwater detention and conveyance systems for commercial and municipal clients
- Led H&H analyses, CLOMR/LOMR applications, and urban floodway re-delineations
- Major projects: Alum Creek and Scioto Mile floodway analyses; The Limited Distribution Center; New Albany Country Club; Easton Market
- Expert witness services: *OH DOT vs. Price, Huspen vs. Cooper (2007)*

Founder & Principal Engineer (*concurrent with ISWS, Ohio/Maryland only*) BaseFlood Engineering, PLLC – Ohio and Maryland *June 2024 – Present*

- Provides water resources consulting, floodplain permitting, and expert witness services in Ohio and Maryland
- Operates independently to maintain ethical separation from Illinois public-sector responsibilities

Professional Certifications and Licensure

Credential	Organization	License/Number
Professional Engineer (P.E.) – Illinois	IDFPR	#062-061996
Professional Engineer (P.E.) – Ohio	Ohio SEBS	#63812
Professional Engineer (P.E.) – Maryland	MDBPIE	#34847
Certified Floodplain Manager (CFM) – National	ASFPM	#US-01-00269
Certified Floodplain Manager (CFM) – Illinois	IAFSM	#IL-11-00606
Board Certified Water Resources Engineer (BC.WRE)	ASCE	#00808

Professional Development and Training

Leading Edge Leadership Certificate Program Parkland College, Champaign, IL |
January–December 2023

- 44 hours of professional development coursework
- Required courses: Coaching for Success, Communicating for Teamwork, DiSC Behavioral Assessment, Authentic Leadership, Feedback That Works
- Elective courses: Delegation, Recognition, Leading Change, Conflict Mastery, Prioritization, Email Control, Time Management

FEMA TMAC Training and Participation Technical Mapping Advisory Council, FEMA |
2024–2025

AI Governance and Implementation - Developed Beta AI Governance Policy for CHAMP (2023–2024) - Developed Prompt Library for H&H review applications - Completed continuing education and self-directed learning on LLM tools (ChatGPT, Claude, Gemini)

[NEEDS INPUT — list any other professional development courses, FEMA training, continuing education, or professional conferences attended]

2. Publications and Scholarship

Peer-Reviewed and Refereed Publications

[NEEDS INPUT — confirm peer-reviewed status and full citation strings for all publications listed below. Identify any additional peer-reviewed publications or proceedings not yet captured.]

- Heistand, G.N. [and co-authors]. (2024). *An Assessment of the Impacts of Climate Change on Cultural Heritage in Illinois*. Illinois State Archaeological Survey/Prairie Research Institute, University of Illinois at Urbana-Champaign.
- [Contributing author]. (2022). *Illinois State Water Plan*. Illinois Department of Natural Resources, Springfield, IL.
- [Lead author or contributing author — confirm]. (2019). *Structure-Specific Flood Risk Assessment Studies*. CHAMP/Illinois State Water Survey, Prairie Research Institute.
- [Lead author]. (2017). *StormStore Feasibility Report*. Illinois State Water Survey, Prairie Research Institute.
- [Lead author]. (2017). *Stormwater Volume Trading Analysis*. Illinois State Water Survey, Prairie Research Institute.

- [Lead author]. (2015). *Urban Flooding Awareness Act Report*. Illinois State Water Survey, Prairie Research Institute.
- Heistand, G.N. (2013). *The Conundrum of Unsteady Flow in a Regulatory Setting*. Proceedings of the 2013 Association of State Floodplain Managers (ASFPM) Annual Conference.

[NEEDS INPUT — are there additional technical publications, proceedings, or co-authored reports from ISWS not listed here?]

Service-Program Deliverables

As Head of CHAMP and former Senior Hydraulic Engineer, Glenn has served as director, technical lead, or reviewer on over **34 FEMA-compliant flood study deliverables** across 25+ Illinois counties, including:

- Flood Insurance Study (FIS) reports and Flood Insurance Rate Maps (FIRMs)
- Hydrologic and Hydraulic (H&H) engineering models (HEC-HMS, HEC-RAS 1D/2D)
- Digital Flood Insurance Rate Map (DFIRM) databases
- Risk MAP products: Flood Risk Reports, Flood Risk Databases, Changes Since Last FIRM maps

Counties/study areas include (representative list): **[NEEDS INPUT — Glenn to provide representative county list if desired; can pull from CHAMP-PM study areas database]**

Glenn directed or co-directed all CHAMP study deliverables submitted to FEMA under cooperative agreements DHS EMC-2021-CA-00012, DHS EMC-2022-CA-00011, DHS EMC-2023-CA-05003, and DHS EMC-2024-CA-05002.

Awards and Recognition

- **PRI Outstanding Collaboration Award**, Prairie Research Institute (2023)
- **Leading Edge Leadership Certificate**, Parkland College (2023)
- **FEMA TMAC Appointment** — selected as Illinois representative to the Technical Mapping Advisory Council (2024)
- **DHS-CISA Levee Subsector Council Appointment** (2024)

[NEEDS INPUT — any other awards, recognition, or peer nominations?]

Citation Impact

[NEEDS INPUT — if Glenn has a Google Scholar profile or ResearchGate, provide citation count and h-index. If not applicable for service-oriented technical reports, note that instead.]

3. Mentoring and Supervision

Supervision and Training of Staff

Glenn directly supervises and supports the professional development of approximately **27 CHAMP staff members**, including:

- 3 Associate Research Scientists (program-level staff)
- 10 Assistant Research Scientists (project engineers and technical leads)
- 6 Senior Scientific Specialists (GIS and outreach professionals)
- 6 Scientific Specialists (engineering and GIS support)
- Academic hourly, student hourly, and extra help staff

He conducts annual performance evaluations, guides merit decisions, leads professional development conversations, and has built a retention-focused team culture over five years of section leadership.

Over his 16-year tenure at ISWS, Glenn has mentored numerous early-career engineers and GIS professionals. Notable staff development initiatives include:

- Established and led the **CHAMP AI Workgroup** — a structured internal program training staff on responsible AI tool use for engineering applications
- Conducted internal technical training on FEMA standards, HEC-RAS 1D/2D modeling, and FEMA's Risk MAP workflow
- Supported multiple CHAMP staff through PRI promotion processes (including contributing to the 2025 promotion packages for Samikshya Pantha and Zhenxing Zhang's section context)

[NEEDS INPUT — list any student hourly workers, graduate students, or undergraduate researchers Glenn has supervised; UIUC affiliate appointments involving student mentoring]

Undergraduate and Graduate Mentorship

- Affiliate Researcher, Water Resources Engineering and Science (WRES), Grainger College of Engineering, UIUC — facilitates applied research collaboration between ISWS scientists and UIUC faculty/students

[NEEDS INPUT — has Glenn served on any graduate student advisory committees or mentored graduate students through his UIUC affiliate appointment?]

Courses, Lectures, and Seminars

- Presented at **CHAMP Annual Retreat** (multiple years) on FEMA program updates, AI adoption, and technical methods
 - Delivered technical training workshops for CHAMP staff and IDNR partners on H&H modeling and FEMA compliance
 - **[NEEDS INPUT — has Glenn co-taught or guest-lectured in any UIUC courses? Any formal workshops or short courses outside ISWS?]**
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4. Leadership

Administrative and Programmatic Leadership Roles

Head, Coordinated Hazard Assessment and Mapping Program (CHAMP) — ISWS | April 2021 – Present - Appointed to lead Illinois' premier FEMA CTP program; all program operations, personnel, and federal deliverables report through this role - Member, ISWS Senior Leadership Team

Co-Chair, Illinois Integrated Water Information Center (IIWIC) Initiative | Ongoing - Joint ISWS/IDNR initiative to build a statewide water data integration strategy - Represents ISWS in interagency planning and governance of Illinois water information infrastructure

Member, Illinois State Water Plan Task Force | Contributed to the 2022 Illinois State Water Plan

Appointed Member, FEMA Technical Mapping Advisory Council (TMAC) | 2024–2025 - One of ~20 national appointees advising the FEMA Administrator on NFIP and national flood mapping policy - Illinois/CTP program representative on a federal advisory committee

Member, DHS-CISA Levee Subsector Council | 2024–2025 - Appointed to the critical infrastructure levee sector advisory body under the Department of Homeland Security

Member, PRI AI Research Frontier Committee | Ongoing - Guides institute-level AI adoption strategy; represents CHAMP's applied AI implementation experience

Project Management Experience

- **PI / Program Director, CHAMP** — Full grant portfolio (~\$15.4M active): proposal development, budget management, federal reporting, deliverable oversight, and coordination with FEMA Region 5 and IDNR

- **Led H&H studies and FEMA deliverables for 25+ Illinois counties** (2009–present)
- **Secured stable CHAMP program funding through FFY2028** — ensured uninterrupted operations through successive grant award cycles

Collaborative Initiatives

- IIWIC Co-Chair (ISWS & IDNR joint initiative)
- FEMA TMAC — represents CTP programs nationally; collaborates with FEMA, NOAA, USGS, and state mapping partners
- ASFPM — national policy and standards collaboration with 50-state network of floodplain managers
- Affiliate Researcher, UIUC WRES group — bridges applied program work and academic research

5. External Funding

Active Grants (as of June 2026)

Glenn serves as Principal Investigator / Program Director for all CHAMP cooperative agreements:

Grant	Funder	Period	Budget
DHS EMC-2023-CA-05003	FEMA/DHS	Sep 2023 – Oct 2027	\$4,908,602
DHS EMC-2022-CA-00011	FEMA/DHS	Sep 2022 – Aug 2026	\$3,303,700
DHS EMC-2021-CA-00012	FEMA/DHS	Sep 2021 – Sep 2026	\$3,250,000
DHS EMC-2024-CA-05002	FEMA/DHS	Sep 2024 – Sep 2028	\$3,241,000
DHS EMC-2024-CA-05001	FEMA/DHS	Sep 2024 – Sep 2026	\$200,000
IDNR OWR-454 (SAFR, Model, Inundation)	State of Illinois/ IDNR	Feb 2025 – Dec 2026	~\$295,000
SBC Jersey County / DHS Sub Lake County	Federal sub-awards	2025–2027	~\$96,000

Grant	Funder	Period	Budget
103 Indirect Cost Recovery – PI Allotment	ISWS Internal	FY2026	\$63,663
Total Active			~\$15.4M

Historical Funding Context

Glenn has overseen CHAMP's evolution from a smaller program to a \$15M+ operation over his tenure. Cumulative grant funding administered under his leadership since 2021 exceeds \$21 million when including completed award cycles.

[NEEDS INPUT — confirm complete list of closed/completed grants from prior award cycles; total cumulative value for CV completeness]

6. Service and Core Research

Professional Society Leadership

Association of State Floodplain Managers (ASFPM) - Co-Chair, Mapping and Engineering Standards Committee | 2025 – Present - Chair | 2022 – 2024 - Secretary | 2021 – 2022 - Treasurer | 2019 – 2021 - [Prior roles: confirm earlier involvement]

Illinois Association for Floodplain and Stormwater Management (IAFSM) - Past Chair | 2019 – 2021 - Chair | 2017 – 2019 - Vice Chair | 2015 – 2017 - Treasurer | 2013 – 2015 - Secretary | 2011 – 2013

Ohio Floodplain Management Association - Board Member | 2008 – 2009

Federal Appointments and Committee Service

- **FEMA Technical Mapping Advisory Council (TMAC)** — Appointed Member | 2024 – 2025
- **DHS-CISA Levee Subsector Council** — Member | 2024 – 2025

Institutional Service

PRI and ISWS Committees: - Promotion Review Committee, ISWS — Member - Data Management Committee, ISWS — Member - Diversity, Equity & Inclusion Committee, ISWS — Member - AI Research Frontier Committee, PRI — Member

Interagency: - Illinois Integrated Water Information Center (IIWIC) — Co-Chair (ISWS/IDNR joint initiative) - Illinois State Water Plan Task Force — Member (contributed to 2022 State Water Plan)

[NEEDS INPUT — any other committee roles, review panel service, grant proposal reviews, manuscript reviews?]

Core Research and Program Activities

Glenn's core technical research and service focuses on applied flood risk science and regulatory flood mapping for the state of Illinois. His work directly supports FEMA's National Flood Insurance Program (NFIP) by:

- Directing the production of updated Flood Insurance Rate Maps (FIRMs) for Illinois communities, incorporating updated topographic data (LiDAR), updated hydrologic and hydraulic analyses, and modern risk communication products
 - Advancing the integration of "Future Risk" data alongside regulatory flood maps, allowing communities to see not just current FEMA-designated flood zones but modeled future-condition flood risk under climate change and development scenarios
 - Developing and implementing AI-assisted workflows for H&H analysis review, program tracking, and regulatory compliance documentation within CHAMP
 - Leading state-level coordination with IDNR on flood risk data management and the IIWIC statewide data infrastructure initiative
 - Serving as Illinois' liaison to FEMA Region 5 and contributing to national CTP program coordination through ASFPM and FEMA TMAC
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7. Engagement and Outreach

Academic and Professional Presentations

National Conference Presentations (Selected):

- **Keynote Address:** *Third Party Mapping and Other National Flood Topics*. Connecticut Association of Flood Managers (CAFM) Annual Conference, Hartford, CT. November 20, 2025.
- **Invited Panelist:** *Navigating the Future of Flood Mapping: A Hybrid Model for a Fuller View of Risk*. Fathom Webinar. October 29, 2025.
- **Invited Panelist:** *Flood Risk Data vs. Regulatory Maps*. NAFSMA Webinar. October 8, 2025.

[NEEDS INPUT — Glenn should provide a complete list of conference presentations, ASFPM Annual Conference presentations, IAFSM presentations, FEMA TMAC public meetings, and any other speaking engagements. Include year, title, event, and location.]

Workshops

- Led internal CHAMP technical training workshops on HEC-RAS, FEMA standards, and AI tool use (multiple years)
- Facilitated CHAMP Annual Retreat sessions on program strategy, AI adoption, and technical development
- **[NEEDS INPUT — any FEMA-sponsored workshops, ASFPM chapter workshops, IAFSM workshops?]**

Public Outreach

- **ISWS Open House** — represented CHAMP with public-facing flood risk mapping exhibits
- **FEMA TMAC Public Meetings** — represented ISWS/CHAMP and engaged with stakeholders on national flood mapping policy
- Stakeholder engagement with Illinois county and municipal officials on FEMA Risk MAP projects, including project initiation meetings, flood map revision outreach, and community appeals support
- Media and public communication on flood risk topics as CHAMP Section Head

Social Media and Web Presence

[NEEDS INPUT — any CHAMP or ISWS social media contributions, web content, or newsletter articles Glenn has authored?]

Educational Materials

- **AI Governance Policy for CHAMP** (2023–2024) — developed internal policy and reference materials for responsible AI use in engineering practice
- **Prompt Library for H&H Review** — created reference library for structured AI prompting in flood study review workflows
- **Custom GPT Expert System** — developed internal GPT assistant for querying FEMA Guidelines & Specifications, supporting staff efficiency and compliance

Appendix: Technical Skills

Category	Tools / Methods
H&H Modeling	HEC-RAS (1D/2D), HEC-HMS, SWMM, TR-55, HY-8
Civil/Stormwater Design	Flowmaster, StormCAD, PondPack

Category	Tools / Methods
GIS/CAD	ArcGIS Pro, ArcMap, AutoCAD
AI and Automation	ChatGPT (GPT-4o), Claude, Gemini, Python scripting
FEMA/Regulatory	NFIP compliance, CLOMR/LOMR, FIS/FIRM production, Risk MAP
Productivity	Microsoft Office (Excel advanced), CHAMP-PM (custom grant/staff tracking), Wrangler/Cloudflare D1

Flags and Items Requiring Input Before Submission

#	Item	Notes
1	ORCID number	Add if applicable
2	Full citation strings for all publications	Confirm authorship order, journal/proceedings name, volume, page, DOI
3	Peer-reviewed vs. technical report distinction	Some items may be refereed publications; others service deliverables
4	Additional publications and reports	Any items not yet listed above
5	Google Scholar / citation count	If profile exists, provide h-index and total citations
6	Study area/county list for deliverables	Optional but useful for quantifying output
7	Complete ASFPM/IAFSM timeline	Confirm early roles; list any national committee work not mentioned
8	Conference presentation list	Full list with year, title, event, location
9	UIUC affiliate research activities	Any specific collaborative projects, co-authored work with UIUC faculty
10	Graduate/student mentorship	Any graduate committee service, student research supervision
11	Manuscript/proposal reviews	Any peer review service not mentioned
12	Social media / web content contributions	CHAMP or ISWS-branded digital content
13	Complete historical grant list	Pre-2021 involvement in CHAMP grants as Senior Hydraulic Engineer